

01 OPTIMIZATION

Gradient descent

$$w_{t+1} = w_t - \eta_t \nabla f(w_t).$$

For quadratic $f(w) = \frac{1}{2} w^T H w - b^T w$, $H \succ 0$, let $L = \lambda_{\max}(H)$.

COND $0 < \eta < 2/L$ gives convergence to the unique minimizer. $\eta = 1/L$ is a safe choice.

! For this quadratic, $0 < \eta < 2/L$ also decreases the objective; $1/L$ is a common sufficient bound for general smooth objectives.

EX $f(w) = 2w^2$: $L = 4$; $w_{t+1} = (1 - 4\eta)w_t$. At $\eta = 1/2$, oscillation persists.

Convexity

$$f(y) \geq f(x) + \nabla f(x)^T (y - x).$$

$\nabla^2 f \geq 0$ implies convexity on a convex domain; $\nabla^2 f \geq \mu I$, $\mu > 0$, implies strong convexity.

COND Differentiable convex f : $\nabla f(w) = 0$ implies a global minimum.

! Stationary point alone is insufficient for a non-convex objective.

Newton / stochastic update

$$w^+ = w - [\nabla^2 f(w)]^{-1} \nabla f(w).$$

Solve $Hp = g$ then set $w^+ = w - p$; avoid forming the inverse. SGD uses a sample or minibatch gradient.

! Newton needs a suitable Hessian and often damping; SGD gradients fluctuate.

02 DERIVATIVE TOOLKIT

For column vectors and symmetric A :

$$\nabla_w (a^T w) = a,$$

$$\nabla_w (w^T A w) = 2Aw,$$

$$\nabla_w \frac{1}{2} \|Xw - y\|^2 = X^T (Xw - y).$$

If A is not symmetric, gradient is $(A + A^T)w$.

Chain rule $z = Ax + b$, $L = L(z)$:

$$\nabla_x L = A^T \nabla_z L.$$

COND Check dimensions: $X \in \mathbb{R}^{n \times d}$, $w \in \mathbb{R}^d$, $y \in \mathbb{R}^n$.

EX $z = 3w + 1$, $L = z^2$: $dL/dw = 6z$.

! A sum loss and a mean loss differ by n in the gradient and regularization scale.

03 CONSTRAINED OPTIMIZATION

For min $f(w)$ subject to $g_i(w) \leq 0$ and

$h_j(w) = 0$:

$$\mathcal{L} = f + \sum_i \alpha_i g_i + \sum_j \nu_j h_j.$$

KKT checklist Stationarity $\nabla_w \mathcal{L} = 0$; primal feasibility; dual feasibility $\alpha_i \geq 0$; complementary slackness $\alpha_i g_i = 0$.

COND Convex problem + suitable constraint qualification (e.g. Slater) gives the usual optimality/strong-duality result.

! Active constraint does not force a strictly positive multiplier; $\alpha_i > 0$ does force activity.

EX min x^2 with $x \geq 1$: $g = 1 - x$, $x^* = 1$, $\alpha^* = 2$.

04 PROXIMAL / NUMERICAL CHECKS

Soft threshold For $\min_u \frac{1}{2} \|u - v\|^2 + \tau \|u\|_1$,

$$u_j = \text{sign}(v_j) \max(|v_j| - \tau, 0).$$

COND Proximal gradient for $g(w) + \lambda \|w\|_1$ uses $v = w - \eta \nabla g(w)$ and $\tau = \eta \lambda$.

EX $v = (2, -4)$, $\tau = .5$: $u = (1.5, 0)$.

! Threshold depends on the step size; do not use λ alone.

Gradient check For a coordinate direction e_j :

$$g_j \approx \frac{f(w + he_j) - f(w - he_j)}{2h}.$$

Compare analytical and numerical gradients away from nondifferentiable points; use double precision and several step sizes.

! Too small h amplifies rounding error; too large h gives truncation error.

Stopping rule Monitor objective, gradient norm, and relative parameter change. Report tolerance and maximum iterations.

COND Small steps alone can reflect a small learning rate rather than proximity to an optimum.

05 LINEAR REGRESSION

$$J(w) = \frac{1}{2n} \|Xw - y\|^2.$$

$$\nabla J = \frac{1}{n} X^T (Xw - y).$$

Normal equation $X^T X w = X^T y$. Full column rank gives a unique least-squares solution.

Otherwise use a pseudoinverse or regularization.

COND An intercept is a column of ones; check whether it is penalized.

EX $X = (1, 2)^T$, $y = (2, 4)^T$: $w^* = 2$.

! In code, use least-squares/QR/SVD solvers instead of computing $(X^T X)^{-1}$.

Residuals $r = y - X\hat{w}$; at an unregularized least-squares optimum, $X^T r = 0$.

06 REGULARIZATION

Ridge

$$J = \frac{1}{2n} \|Xw - y\|^2 + \frac{\lambda}{2} \|w\|^2.$$

$$(X^T X + n\lambda I)w = X^T y.$$

COND $\lambda > 0$ makes this matrix positive definite. Standardize features before comparing penalty strength.

! If data loss has no $1/n$, the normal equation has λI under the matching $1/2$ convention.

Lasso $J = \frac{1}{2n} \|Xw - y\|^2 + \lambda \|w\|_1$. For $w_j = 0$, the subgradient is $[-1, 1]$; lasso can set coefficients exactly to zero.

Bias / variance For squared prediction error at fixed x :

$$\mathbb{E}[(Y - \hat{f}(x))^2] = \sigma^2 + \text{Bias}^2 + \text{Var}(\hat{f}(x)).$$

COND Expectation is over training sets and independent test noise; $\hat{f}(x) = \mathbb{E}[Y | x]$.

! Stronger regularization often raises bias and lowers variance; this is a tendency, not a universal monotonic law.

07 LOGISTIC REGRESSION

$y \in \{0, 1\}$; $z = w^T x + b$:

$$p = \sigma(z) = \frac{1}{1 + e^{-z}}, \quad \sigma' = p(1 - p).$$

$$\ell = -y \log p - (1 - y) \log(1 - p).$$

$$\partial \ell / \partial z = p - y, \quad \nabla_w \ell = (p - y)x.$$

Stable logit loss

$$\ell = \max(z, 0) - yz + \log(1 + e^{-|z|}).$$

COND Decision threshold $p \geq \tau$ is $z \geq \log \frac{\tau}{1-\tau}$ for $0 < \tau < 1$.

EX $z = 0$, $y = 1$: $p = 0.5$, logit gradient $= -0.5$.

! Logits are unrestricted real numbers; probabilities lie in $[0, 1]$. Threshold 0.5 is a choice.

Multiclass $p_k = e^{z_k} / \sum_j e^{z_j}$,

$\ell = -\sum_k y_k \log p_k$; $\partial \ell / \partial z_k = p_k - y_k$ for a target distribution with $\sum_k y_k = 1$.

! Subtract $\max_j z_j$ before exponentiating.

08 LINEAR MODEL DECISIONS

Perceptron If $y_i(w^T x_i + b) \leq 0$, update

$$w \leftarrow w + \eta y_i x_i, \quad b \leftarrow b + \eta y_i.$$

COND Labels $\{-1, +1\}$. Classical finite-mistake convergence needs separability and bounded inputs.

! A perceptron does not generally maximize the margin or produce calibrated probabilities.

MAP interpretation A Gaussian prior gives an L_2 penalty; a Laplace prior gives an L_1 penalty, with scale matched to the likelihood convention.

Feature transforms Polynomial features allow nonlinear boundaries with a model still linear in its parameters. Add interactions only using training/validation evidence.

EX Features $(1, x, x^2)$: prediction $w_0 + w_1 x + w_2 x^2$.

! Linear in parameters does not mean linear in the original input.

Cost-sensitive threshold For calibrated

$p = P(y = 1 | x)$ and only false-positive/false-negative costs:

$$\tau = \frac{C_{FP}}{C_{FP} + C_{FN}}.$$

COND Both costs nonnegative with positive sum; predict positive when $p \geq \tau$.

! A changed class prior can invalidate probability calibration and the chosen threshold.

09 SVM / MARGINS

$y_i \in \{-1, +1\}$; $f_i = w^T x_i + b$.

$$\gamma_i = \frac{y_i f_i}{\|w\|}.$$

Hard margin

$$\min_{w,b} \frac{1}{2} \|w\|^2 \quad \text{s.t. } y_i f_i \geq 1.$$

Canonical margin width is $2/\|w\|$.

COND Hard margin requires linear separability in the chosen feature space.

! Positive rescaling of (w, b) changes functional margin, while geometric margin is unchanged.

Soft margin

$$\min \frac{1}{2} \|w\|^2 + C \sum_i \xi_i, \quad \xi_i \geq 0,$$

$y_i f_i \geq 1 - \xi_i$. Equivalently use hinge loss

$\max(0, 1 - y_i f_i)$.

Dual / KKT $0 \leq \alpha_i \leq C$, $\sum_i \alpha_i y_i = 0$;

$w = \sum_i \alpha_i y_i \phi(x_i)$.

COND $0 < \alpha_i < C \Rightarrow y_i f_i = 1$; $\alpha_i = C$ allows points on/inside the margin.

! $\alpha_i = C$ does not imply misclassification; $\alpha_i = 0$ can occur on the margin in degenerate cases.

EX $y_i f_i = 0.3$: hinge loss = 0.7; prediction has the correct sign but lies inside the margin.

10 KERNELS

$$K(x, z) = \phi(x)^T \phi(z).$$

Validity Symmetric kernel; every finite Gram matrix $G_{ij} = K(x_i, x_j)$ must be positive

semidefinite:

$$c^T G c \geq 0 \quad \text{for all } c.$$

Common forms Linear $x^T z$; polynomial

$(x^T z + c)^p$ for integer $p \geq 1$, $c \geq 0$; RBF

$e^{-\gamma \|x - z\|^2}$, $\gamma > 0$.

COND Nonnegative sums and pointwise products of PSD kernels remain PSD.

! A symmetric similarity function alone need not be a valid kernel.

EX $G = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}$ has eigenvalue -1 : fails PSD.

Prediction

$$f(x) = \sum_i \alpha_i y_i K(x_i, x) + b.$$

Only nonzero coefficients contribute; no explicit ϕ is needed.

11 DISTANCE / CLUSTERING

k-means

$$\min_{\mu, a} \sum_i \|x_i - \mu_{a_i}\|^2.$$

Alternate nearest-centroid assignment and mean updates; objective does not increase in either exact step.

COND Squared Euclidean distance; fix the treatment of empty clusters. Use multiple initializations.

! Convergence is to a local solution; cluster IDs have no intrinsic order.

EX Points 0, 2, 9, 11, $k = 2$: groups $\{0, 2\}$, $\{9, 11\}$ have means 1, 10.

Cosine similarity

$$s(x, z) = \frac{x^T z}{\|x\| \|z\|}.$$

COND Both vectors nonzero. Normalize first when direction rather than magnitude is relevant.

! Cosine similarity is not Euclidean distance; a zero vector needs an explicit convention.

12 DECISION TREES

Class fractions p_k in a node:

$$G = 1 - \sum_k p_k^2, \quad H = -\sum_k p_k \log p_k.$$

Use $0 \log 0 = 0$. For a split into L, R :

$$\Delta I = I(P) - \frac{n_L}{n} I(L) - \frac{n_R}{n} I(R).$$

COND Choose the split maximizing impurity reduction; for regression, minimize weighted squared error.

EX Binary node (0.5, 0.5): Gini = 0.5; pure children give gain 0.5.

! Weight child impurities by their sample counts; averaging them equally is generally wrong.

Capacity control Depth limit, minimum leaf size, pruning. Deep trees fit interactions but may have high variance.

COND Axis-aligned splits do not require feature standardization.

13 BAGGING / RANDOM FORESTS

Train models on bootstrap samples; average regression predictions or vote for classification.

$$\text{Var}(\bar{h}) = \sigma^2 \left(\rho + \frac{1 - \rho}{B} \right).$$

COND Formula assumes equal individual variance and pairwise correlation ρ .

Random forests additionally sample candidate features at each split to reduce correlation.

Out-of-bag Evaluate each observation using only trees whose bootstrap samples excluded it.

! Many highly correlated trees retain a variance floor; B alone cannot remove $\rho \sigma^2$.

EX $\rho = 0$, $B = 10$: variance $\sigma^2/10$; $\rho = 0.5$: $0.55\sigma^2$.

14 BOOSTING

Gradient boosting

$$r_{im} = - \left. \frac{\partial \ell(y_i, F)}{\partial F} \right|_{F=F_{m-1}(x_i)}.$$

Fit h_m to pseudo-residuals; update

$$F_m = F_{m-1} + \eta \gamma_m h_m.$$

EX Squared loss $\ell = \frac{1}{2}(y - F)^2$: pseudo-residual = $y - F$.

AdaBoost $\epsilon_m = \sum_i q_i \mathbf{1}[y_i \neq h_m(x_i)]$;

$$a_m = \frac{1}{2} \log \frac{1 - \epsilon_m}{\epsilon_m}; q_i^+ \propto q_i e^{-a_m y_i h_m(x_i)}.$$

COND Binary h_m , $y_i \in \{-1, +1\}$; the standard positive-weight step needs $0 < \epsilon_m < 1/2$.

! Boosting fits models sequentially. Noisy labels can receive large weight.

15 ENSEMBLE DECISIONS

Aggregation Regression: mean of predictions.

Classification: vote or average aligned class probabilities, then apply the decision rule.

COND Align class order before averaging probabilities from different models.

! Averaging logits and averaging probabilities give different ensembles.

Stacking Train a second-level model on out-of-fold base-model predictions; refit base models on full training data for deployment.

COND All folds and feature processing obey the same leakage constraints as ordinary CV.

! Training the stacker on in-sample predictions gives it an overly optimistic view of base models.

EX Two binary probabilities .2, .8: probability average = .5. Agreement and confidence remain separate questions.

Choose / check Tune tree capacity and ensemble size with validation; record metric, split, seed, and hyperparameters.

16 SVM DUAL / WORKED CHECK

$$\begin{aligned} \max_{\alpha} \quad & \sum_i \alpha_i \\ & - \frac{1}{2} \sum_{i,j} \alpha_i \alpha_j y_i y_j \\ & \quad \cdot K(x_i, x_j) \\ \text{s.t.} \quad & 0 \leq \alpha_i \leq C, \\ & \sum_i \alpha_i y_i = 0. \end{aligned}$$

COND This form assumes an unpenalized bias. Recover $b = y_i - \sum_j \alpha_j y_j K(x_j, x_i)$ from any $0 < \alpha_i < C$.

EX Linear hard-margin sample: $(x_1, y_1) = (-1, -1)$, $(x_2, y_2) = (1, +1)$. $w = 1$, $b = 0$, $\alpha_1 = \alpha_2 = 1/2$; margin width = 2.

! If no coefficient is strictly between 0 and C , a free support-vector formula cannot be used directly; use KKT bounds for b .

17 PROBABILITY / BAYES

$P(A \mid B) = \frac{P(B \mid A)P(A)}{P(B)}$, $P(B) > 0$.

Total probability For a disjoint, exhaustive partition $\{A_k\}$:

$$P(B) = \sum_k P(B \mid A_k)P(A_k).$$

EX Prevalence 0.01, sensitivity 0.9, false-positive rate 0.05:

$$P(D \mid +) = \frac{.9(.01)}{.9(.01) + .05(.99)} \approx .154.$$

! Sensitivity $P(+ \mid D)$ differs from the posterior $P(D \mid +)$. Include false positives in the denominator.

Expectation / variance

$$\mathbb{E}[aX + b] = a\mathbb{E}[X] + b;$$
$$\text{Var}(aX + b) = a^2 \text{Var}(X).$$
$$\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y) + 2\text{Cov}(X, Y).$$

COND Independence implies zero covariance when moments exist. The converse generally fails.

18 CLASSIFICATION METRICS

$$\text{Prec} = \frac{TP}{TP + FP}, \quad \text{Rec} = \frac{TP}{TP + FN}.$$
$$F_1 = \frac{2TP}{2TP + FP + FN}.$$

Specificity = $TN / (TN + FP)$; accuracy = $(TP + TN) / N$.

COND Specify the positive class and handle zero denominators explicitly.

! Accuracy can hide failure on a rare class. Macro averaging weights classes equally; micro pooling weights counts.

EX $TP = 8, FP = 2, FN = 4$: precision = .8, recall = $2/3, F_1 = 8/11$.

Thresholds Raising a score threshold cannot increase predicted positives; recall is nonincreasing. Precision need not increase monotonically.

! ROC uses $FPR = FP / (FP + TN)$; a precision-recall curve uses precision.

19 VALIDATION / LEAKAGE

Procedure Split data $\rightarrow$ fit transforms on training data $\rightarrow$ tune on validation/CV $\rightarrow$ evaluate once on held-out test data.

COND Refit preprocessing inside every CV fold. Split by time, person, or group when the task requires it.

! Scaling, feature selection, imputation, and target encoding can all leak held-out information.

Model selection Validation chooses hyperparameters. Test estimates generalization after choices are fixed. Nested CV separates inner tuning from outer evaluation.

EX Repeated patients: keep all records of a patient in the same fold.

! Picking a threshold from test labels is also test-set tuning.

Distance-based models Standardize features when units differ; k -NN stores training data, and prediction depends on distance and k .

COND Fit centering/scaling statistics on the training split only.

20 ESTIMATION / UNCERTAINTY

Sample mean Independent identically distributed X_i with finite variance σ^2 :

$$\mathbb{E}[\bar{X}] = \mu, \quad \text{Var}(\bar{X}) = \sigma^2/n.$$

Sample variance $s^2 = \sum_i (X_i - \bar{X})^2 / (n - 1)$ is unbiased for $n > 1$.

COND The $1/n$ variance result depends on independence; correlated observations add covariance terms.

Gaussian mean interval Known σ : $\bar{X} \pm z_{1-\alpha/2} \sigma / \sqrt{n}$; unknown σ : use t_{n-1} and s for normal samples.

! A frequentist confidence level describes repeated-sampling coverage.

EX Known $\sigma = 2, n = 100$: standard error = .2, approximate 95% half-width = .392.

Bootstrap Resample observations with replacement; recompute the statistic. Respect groups or time dependence with an appropriate resampling scheme.

! A narrow interval cannot repair selection bias or a mismatched evaluation population.

21 NEURAL NETS / BACKPROP

Layer $a_\ell = W_\ell h_{\ell-1} + b_\ell, h_\ell = \phi(a_\ell)$.

$$\delta_\ell = (W_{\ell+1}^T \delta_{\ell+1}) \odot \phi'(a_\ell).$$
$$\nabla_{W_\ell} L = \delta_\ell h_{\ell-1}^T, \quad \nabla_{b_\ell} L = \delta_\ell.$$

COND These expressions are per example with column activations. Sum/average consistently across a batch.

Activations ReLU $\max(0, a)$: derivative 1 for $a > 0, 0$ for $a < 0$; at 0 choose a subgradient. Sigmoid derivative $p(1 - p)$; tanh derivative $1 - h^2$.

! Apply the activation derivative elementwise; transpose the downstream weight matrix.

EX $h = \text{ReLU}(wx), x = 2, w = 1, L = \frac{1}{2}(h - 3)^2: dL/dw = -2$.

Training / inference Dropout is stochastic during training; batch normalization uses batch statistics during training and stored statistics at inference.

! Switch to evaluation mode for inference.

22 PCA / PROJECTIONS

Center $X \in \mathbb{R}^{n \times d}$ first. Let $X = U\Sigma V^T$ be an SVD.

$$Z = XV_k, \quad \hat{X} = ZV_k^T.$$

Top right singular vectors maximize projected variance and minimize rank- k squared reconstruction error.

Explained variance ratio

$$\text{EVR}_k = \frac{\sum_{j=1}^k \sigma_j^2}{\sum_j \sigma_j^2}.$$

COND Sample covariance is $X^T X / (n - 1)$ for centered observations. Scale first if units should receive comparable influence.

! PCA is unsupervised. High variance need not predict the label; component signs are arbitrary.

EX $X^T X = \text{diag}(9, 1)$: first component along axis 1, EVR = .9.

23 NAIVE BAYES / LIKELIHOOD

$$\hat{y} = \arg \max_c \left[\log P(c) + \sum_j \log P(x_j \mid c) \right].$$

COND Assumes features are conditionally independent given the class. Choose a likelihood matching feature type.

Multinomial smoothing

$$P(j \mid c) = \frac{N_{cj} + \alpha}{\sum_r N_{cr} + \alpha V}.$$

V is vocabulary size; $\alpha > 0$ avoids zero token probabilities.

! Conditional independence is different from unconditional independence. Accumulate logs to avoid underflow.

EX One unseen token makes an unsmoothed product zero; additive smoothing preserves a finite log score.

24 MODEL / LOSS MATCHING

Regression losses Squared error targets a conditional mean; absolute error targets a conditional median, when the relevant expectations exist.

Classification losses Cross-entropy trains a probability model; hinge loss targets a margin. Their numerical values are not directly comparable.

COND Compare models on the same held-out metric and data split.

Learning rate / batch size Larger batches reduce sampling noise under ordinary independent sampling. Learning rate, batch size, and schedule interact.

! More epochs need not improve held-out performance; select early stopping using validation data.

EX Train loss falls while validation loss rises: in-spect capacity, regularization, leakage, and split mismatch.

Reproducibility Record preprocessing, seed, split identities, metric convention, and model settings; preserve a held-out test set.

! A fixed seed is not evidence that evaluation data represent deployment data.

25 ASYMPTOTIC ANALYSIS

Bounds $f = O(g)$: eventual upper bound; $f = \Omega(g)$: lower bound; $f = \Theta(g)$: both. State what n measures.

Common growth

$$1 < \log n < n < n \log n < n^2 < 2^n < n!$$

asymptotically.

Loop patterns Halving/doubling a positive index: $\Theta(\log n)$. Triangular loop $\sum_{i=1}^n i$: $\Theta(n^2)$.

EX Outer loop doubles i ; inner loop runs i times: $1 + 2 + 4 + \dots \leq 2n$, hence $\Theta(n)$.

! Multiply costs only when inner work has the claimed cost on every outer iteration.

Master theorem For $T(n) = aT(n/b) + f(n)$, $a \geq 1, b > 1$, compare f to $n^{\log_b a}$. Polynomially smaller: $\Theta(n^{\log_b a})$; equal: $\Theta(n^{\log_b a} \log n)$; polynomially larger plus regularity: $\Theta(f(n))$.

COND The third case needs $af(n/b) \leq cf(n)$ eventually for some $c < 1$.

! These three basic cases do not cover every recurrence.

26 SEARCH / SORT

Binary search invariant Search $[l, r)$; $m = l + \lfloor (r - l) / 2 \rfloor$. Lower bound finds the first $a[m] \geq x$.

- If $a[m] < x$, set $l = m + 1$.
- Otherwise set $r = m$; stop at $l = r$.

COND Array sorted in the comparison order. Result may equal n .

! For membership, also check $l < n$ and $a[l] = x$.

Sort comparison

Method	Time	Stable?
Merge	$\Theta(n \log n)$	yes*
Heap	$O(n \log n)$	no
Quick	avg. $O(n \log n)$	no
Insertion	worst $O(n^2)$	yes*

COND *Standard tie-preserving implementation. Quicksort worst case is $O(n^2)$.

EX Lower bound of 2 in $[1, 2, 2, 4]$ is index 1, with zero-based indexing.

27 DATA STRUCTURES

Choose by operation Stack: LIFO; queue: FIFO; hash map: expected constant-time lookup under suitable hashing/load; balanced BST: $O(\log n)$ lookup/update.

Binary heap Peek $O(1)$; insert/extract $O(\log n)$; bottom-up heap construction $O(n)$.

! A heap is partially ordered; arbitrary-value search can take $O(n)$.

Amortized cost Dynamic array doubling gives amortized $O(1)$ append, although a resizing append costs $O(n)$.

EX n pushes with doubling copy $1 + 2 + 4 + \dots < n$ old elements in total.

! Expected, amortized, and worst-case bounds describe different guarantees.

28 UNION-FIND / PREFIX SUMS

Disjoint sets $\text{find}(x)$ returns a representative; $\text{union}(x, y)$ merges sets. Use path compression plus union by size/rank.

COND A sequence of operations has near-constant amortized cost $O(\alpha(n))$ per operation under the standard implementation.

! The representative is an implementation choice; it is not necessarily the smallest member.

EX Union $(0, 1)$ and $(1, 2)$ makes $\text{find}(0) = \text{find}(2)$ true.

Prefix sum Define $P[0] = 0$, $P[i + 1] = P[i] + a[i]$.

$$\sum_{i=l}^{r-1} a[i] = P[r] - P[l].$$

COND Half-open range $[l, r)$; preprocessing $O(n)$, query $O(1)$ on an unchanged array.

! Use a numeric type large enough for accumulated sums. Point updates invalidate later prefix values.

EX $a = [2, 3, 5], P = [0, 2, 5, 10]$; range $[1, 3)$ has sum $10 - 2 = 8$.

29 GRAPH TRAVERSAL

Let V be vertex count and E edge count; use adjacency lists.

BFS Queue; mark a vertex discovered when enqueueing it. Finds minimum edge-count paths from a source in $O(V + E)$.

COND Unweighted or equal-weight edges for shortest-path interpretation.

! Marking only on dequeue may enqueue the same vertex repeatedly.

DFS Recursion/stack; discovery/finish states. Directed cycle exists when an edge points to a vertex on the active DFS stack.

COND In an undirected simple graph, ignore the parent edge when checking for cycles.

EX $s \rightarrow a, s \rightarrow b, a \rightarrow t$: BFS gives distances $(0, 1, 1, 2)$.

Topological ordering Kahn: enqueue zero-indegree vertices, remove them and decrement neighbor indegrees. Fewer than V outputs implies a directed cycle.

! A topological ordering exists exactly for a directed acyclic graph.

30 SHORTEST PATHS / MST Relaxation

$$d[v] \leftarrow \min(d[v], d[u] + w(u, v)).$$

Initialize source to 0, others to ∞ .

Dijkstra Extract smallest tentative distance; relax outgoing edges. Binary heap: $O((V + E) \log V)$ with decrease-key.

COND All edge weights nonnegative. A lazy heap variant skips stale entries and has an E -dependent heap bound.

! Negative edges can invalidate settling a vertex. Do not add weights to an unreachable/infinite distance.

Bellman-Ford / DAG Bellman-Ford: $V - 1$ passes, $O(VE)$; a further relaxation detects a reachable negative cycle. DAG: topological relaxation, $O(V + E)$, negative edges allowed.

Minimum spanning tree Kruskal: sort edges, union endpoints from different components, $O(E \log E)$. Prim: grow a tree by the cheapest crossing edge.

COND Undirected weighted graph; disconnected input gives a minimum spanning forest.

! An MST minimizes total tree weight; it need not preserve shortest source-to-vertex paths.

31 DYNAMIC PROGRAMMING

Design steps State meaning $\rightarrow$ base cases $\rightarrow$ transition $\rightarrow$ dependency order $\rightarrow$ answer location. Time $\approx$ states $\times$ work/state; memory depends on stored dependencies.

0/1 knapsack

$$D[i, c] = \max\{D[i - 1, c], v_i + D[i - 1, c - w_i]\}.$$

Second term only when $c \geq w_i$; base $D[0, c] = 0$ for optional selection and nonnegative capacity.

COND Each item is usable at most once. One-array implementation updates capacity downward.

! Ascending capacity reuses the current item, producing the unbounded version.

EX One item $(w, v) = (2, 3)$, capacity 4: 0/1 answer = 3; unbounded answer = 6.

Reconstruction Store choices or compare predecessor states. Rolling arrays save space but may remove the trace needed to reconstruct a solution.

32 DP / CODE MICRO-EXAMPLE

0/1 knapsack, one array Initialize $\text{dp}[0 \dots C]$ to zero, then:

```
for (w, v) in items:
    for c = C down to w:
        dp[c] = max(dp[c], v + dp[c - w])
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COND Integer nonnegative capacities; optional item selection. Time $O(nC)$, memory $O(C)$.

! $O(nC)$ is pseudopolynomial because capacity is encoded with $O(\log C)$ bits.

Longest common subsequence Matching characters: $D[i, j] = 1 + D[i - 1, j - 1]$; otherwise $D[i, j] = \max(D[i - 1, j], D[i, j - 1])$.

COND Base row/column zero; time $O(nm)$. A subsequence may skip positions.

! Longest common substring uses a different recurrence and requires contiguous matches.

EX ABC and AC: LCS length 2, common substring length 1.